Validation

Why regularisation, four objectives, one score

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Weekend 1

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Section 1

Why regularisation

The number you can measure and the number you want

- Fitting minimises the miss on the data you hold. Vapnik calls that the **empirical risk**.
- What you are paid for is the miss on the cases you have never seen. Vapnik calls that the **risk**.
- Every method in this weekend minimises the first and hopes it tracks the second.

The UN games shows what happens when it stops tracking. Ordinary least squares on 203 countries and 203 columns scores 0.997 on the countries it fitted and −9.41 on the 51 it did not.

$$\boxed{R_{\text{emp}}(w) = \frac{1}{n}\sum_{i=1}^{n} L(y_i, \hat{y}_i)}$$

what you can compute

$$\Big\downarrow \text{ hope}$$

$$\boxed{R(w) = \int L(y, \hat{y})\, dP(x, y)}$$

what you are paid for

Vapnik's bound: the gap has a price tag

$$\boxed{R(w)} \leq \boxed{R_{\mathrm{emp}}(w)} + \boxed{\Omega\left(\frac{h}{n}\right)}$$

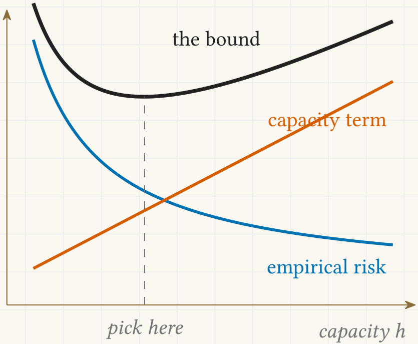
the true risk — *what you drove down* — *the capacity term*

- h is the **capacity** of the family you searched, its VC dimension, and n is how many examples you have. The term grows with h and shrinks with n.
- Driving R_{emp} to zero inside a large enough family buys nothing, because the second term pays for it.

For indicator functions Vapnik gives the term explicitly: with probability at least $1-\eta$, $\Omega = \sqrt{\left(h(\ln \frac{2n}{h} + 1) - \ln \frac{\eta}{4}\right)/n}$.

Structural risk minimisation

- Vapnik's prescription is to stop minimising the first term alone and minimise the **sum**.
- Order the families by capacity, $S_1 \subset S_2 \subset \ldots$ with $h_1 \leq h_2 \leq \ldots$, and pick the family where the sum bottoms out.
- Regularisation is how that is done with a knob instead of a list. One α indexes the whole nested family: large α is a small family, $\alpha \to 0$ is the largest one.



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Section 2

Four objectives

Least squares, the one with no penalty

$$\boxed{\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}$$

no penalty

miss on the data

Minimised over w. With p columns and n rows, a unique minimiser needs $p < n$. Past that the miss can be driven to zero by infinitely many different w, and the solver returns one of them. On all 493 UN columns it scores 1.000 on the training countries and 0.333 on the held back ones.

- This is the whole family. Nothing restricts how large the coefficients may grow, so nothing stops the fit from spending them on noise.

Ridge, the sum of squares

$$\boxed{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} + \boxed{\alpha \sum_{j=1}^{p} w_j^2}$$

miss on the data *the L2 penalty*

Squaring is smooth at zero, so the derivative of the penalty vanishes there and no coefficient is ever pushed exactly to zero. Every column stays in the model with a smaller weight.

- It has a closed form, $\hat{w} = \left(X^\top X + \alpha I\right)^{-1} X^\top y$, which is invertible for any $\alpha > 0$ even when $X^\top X$ is singular. That is the original argument for it.
- On the UN data its best test score is 0.592 at $\alpha = 81.1$, keeping **487 of 493** coefficients.

Lasso, the sum of absolute values

$$\boxed{\sum_{i=1}^{n} (y_i - \hat{y}_i)^2} + \boxed{\alpha \sum_{j=1}^{p} |w_j|}$$

miss on the data *the L1 penalty*

The absolute value has a corner at zero, so the penalty keeps pulling with full force right up to zero and coefficients land exactly on it. The lasso therefore *selects* as well as shrinks.

- No closed form. It is solved numerically, by coordinate descent in scikit-learn.
- On the UN data its best test score is 0.599 at $\alpha = 285$, keeping **109 of 493** coefficients. The other 384 are exactly zero.
- A model that keeps 109 columns is easier to talk about. Weekend 1 spends chapter four of the UN games on why that is not the same as being right about which columns matter.

Elastic net, both penalties at once

$$\boxed{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} + \boxed{\alpha\left(\rho\sum_j |w_j| + \frac{1-\rho}{2}\sum_j w_j^2\right)}$$

miss on the data *the mixed penalty*

One knob α for how much penalty, a second knob ρ for how it is split. $\rho = 1$ is the lasso and $\rho = 0$ is ridge.

- It exists because the lasso behaves badly when columns are strongly correlated: it tends to pick one of a correlated group almost at random and zero the rest. The L2 part pulls the group together so they enter or leave as a block.
- On the UN data, at $\rho = 0.5$, its best test score is 0.592 at $\alpha = 0.811$, keeping **485 of 493** coefficients.

The four side by side

Method	Penalty	Zeroes?	Best test	Kept
Least squares	none	no	0.333	493
Ridge	$\alpha \sum w_j^2$	no	0.592	487
Lasso	$\alpha \sum \lvert w_j \rvert$	yes	0.599	109
Elastic net	both, split by ρ	yes	0.592	485

One trap, and it costs an afternoon

The three α values above are **not on one scale**, because scikit-learn does not normalise the three objectives the same way. Ridge minimises $\|y - Xw\|_2^2 + \alpha\|w\|_2^2$, with no division. Lasso and elastic net both carry a factor $\frac{1}{2n}$ on the miss term. So an α of 81.1 for ridge and 285 for the lasso say nothing about which model is penalised harder. Compare the coefficients that survive, or the scores. Never the raw α.

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Section 3

Reading the score

The R squared score

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}$$

- The numerator is your model's total squared miss.
- The denominator is the miss of the laziest model there is: it ignores every column and predicts the average $\bar{y}$.
- So R^2 asks one question. How much of that lazy model's error did you remove.
- It has no floor, and $\bar{y}$ is the mean of whichever set you score on.

1	*every prediction exact*
0.6	*a good fit on this data*
0	*as good as predicting $\bar{y}$*
< 0	*worse than predicting $\bar{y}$*

-9.41 at $p = n$, and -240.8 at $p = 201$